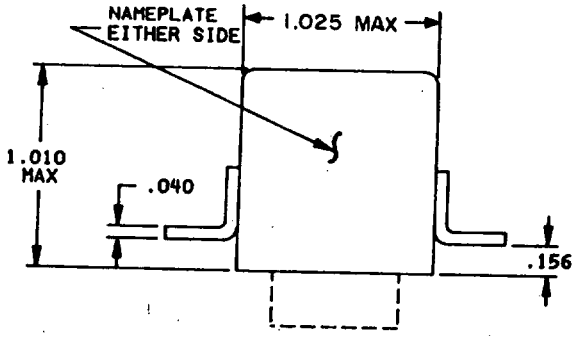
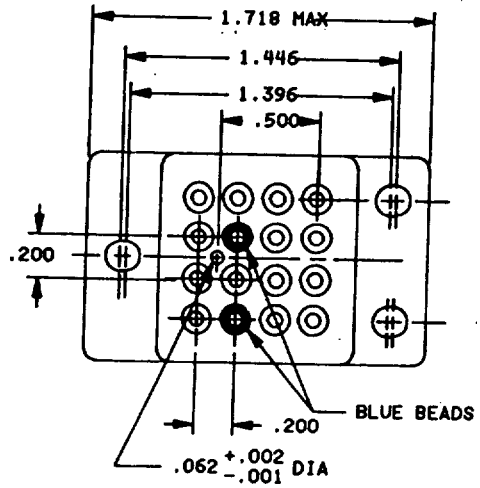
INACTIVE FOR NEW DESIGN AFTER 29 Apr 1988
USE MIL-R-6106/51

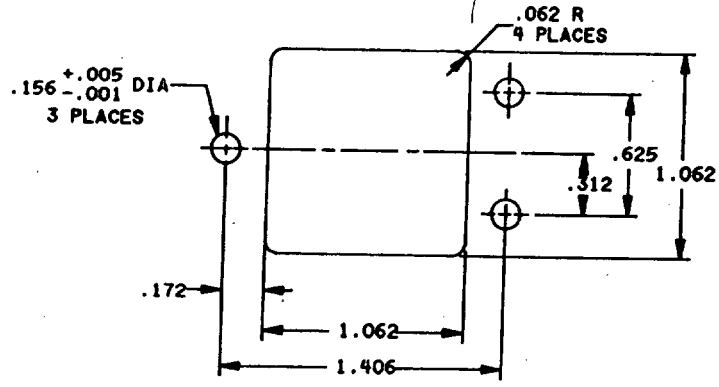
FED. SUP CLASS
5945



RAISED VERTICAL FLANGE MOUNT (-1,-2,-3,-4,&-5)



POLARIZING PIN(FOR-2,-4,-5)
TERMINAL LAYOUT
-1,-2,-3,-4,-5



(MOUNTING HOLE LAYOUT (FOR -1,-3)

FIGURE 1. Outline drawing (for details see tables I and II).

(G) denotes changes

P.A AF-85 Other Cust NAVY-AS ARMY-ER	International interest	TITLE RELAYS, ELECTROMAGNETIC, MAGNETIC LATCH, 10 AMPERES, 4 PDT, ALL WELDED, HERMETICALLY SEALED.	MILITARY STANDARD MS27745
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 1 OF 7

AMSC N/A (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE. 5945-0762-2

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

APPROVED 3 JUNE 71 REVISED (F) 28 MAY 85 (G) 29 APR 88

User activities: Army - Navy - Air Force -

Review activities: Army - Navy - EC Air Force - 99

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:50Z
Check the source to verify that this is the current version before use.

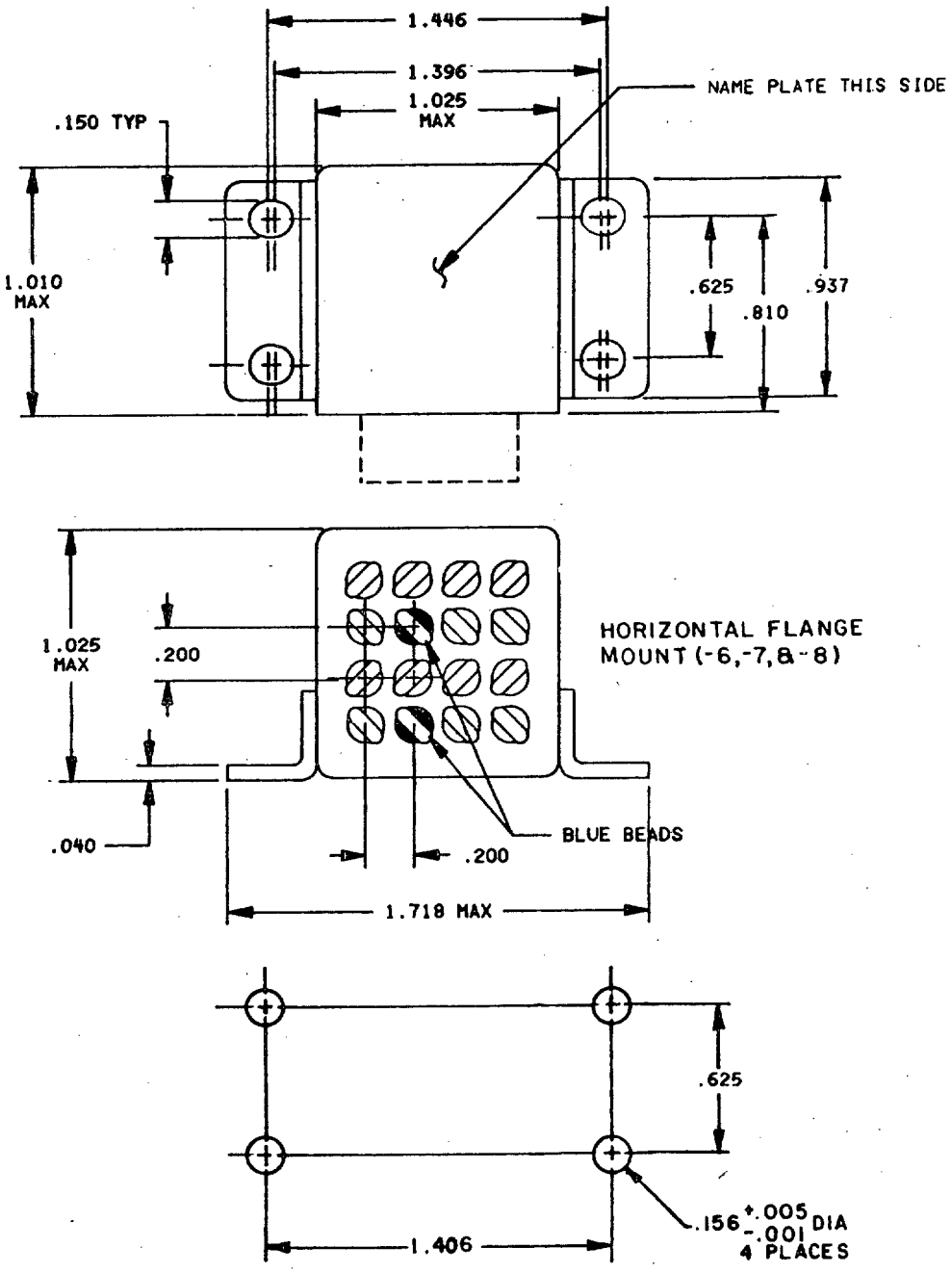


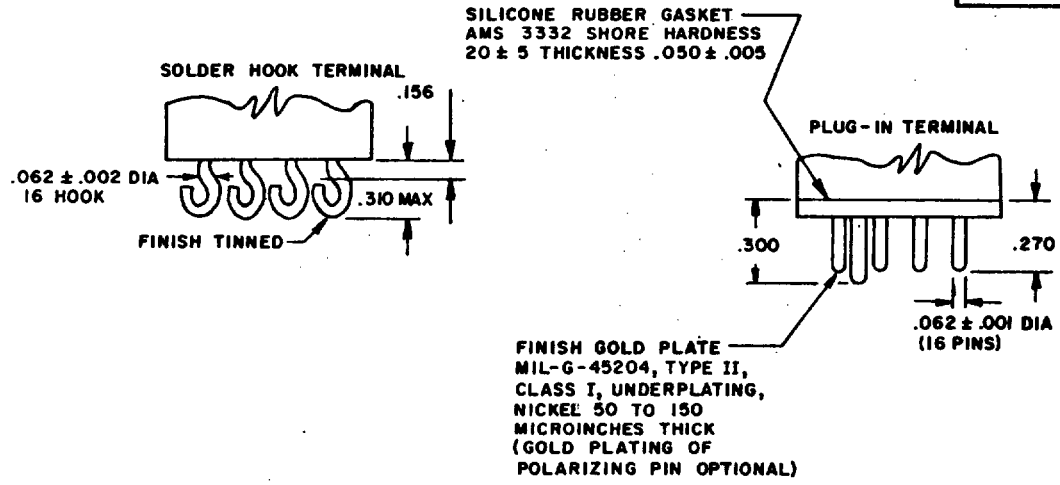
FIGURE 1. Outline drawing (for details see tables I and II) - Continued.

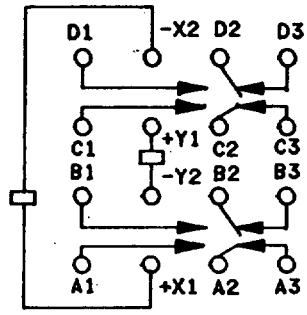
User activities: Army - Navy - Air Force -
Review activities: Army - Navy - EC Air Force - 99
This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

APPROVED 3 JUNE 71
REVISED (G) FOR CHANGES SEE PAGES 1, 6, AND 7

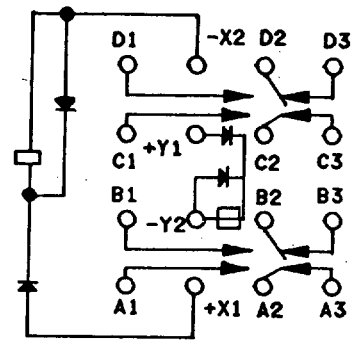
P.A AF-85 Other Cust NAVY-AS ARMY-ER	International Interest	TITLE RELAYS, ELECTROMAGNETIC, MAGNETIC LATCH, 10 AMPERES, 4 PDT, ALL WELDED, HERMETICALLY SEALED.	MILITARY STANDARD MS27745
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 2 OF 7

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:50Z
Check the source to verify that this is the current version before use.





CIRCUIT DIAGRAM
TERMINAL VIEW
-1,-2,-3,-4,-6,AND-7



CIRCUIT DIAGRAM
TERMINAL VIEW
-5 & -8

TO CLOSE NO. 1 CONTACTS (OPERATE)
ENERGIZE X1 AND X2.
TO CLOSE NO. 3 CONTACTS (RESET)
ENERGIZE Y1 AND Y2.

WHEN ENERGIZED WITH POLARITY INDICATED CONTACTS WILL TRANSFER.

INCHES	MM	INCHES	MM
.001	0.03	.310	7.87
.002	0.05	.312	7.92
.005	0.13	.500	12.70
.040	1.02	.625	15.88
.050	1.27	.810	20.57
.062	1.57	.937	23.80
.150	3.81	1.010	25.65
.156	3.96	1.025	26.04
.172	4.37	1.062	26.97
.200	5.08	1.396	35.46
.218	5.54	1.406	35.71
.270	6.86	1.446	36.73
.300	7.62	1.718	43.64

FIGURE 1. Outline drawing (for details see tables I and II) - Continued.

APPROVED 3 JUNE 71 REVISED (G) FOR CHANGES SEE PAGES 1, 6, AND 7

User activities: Army - Navy - Air Force -
Review activities: Army - EC Navy - Air Force - 99

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

P.A AF-85 Other Cust NAVY-AS ARMY-ER	International interest	TITLE RELAYS, ELECTROMAGNETIC, MAGNETIC LATCH, 10 AMPERES, 4 PDT, ALL WELDED, HERMETICALLY SEALED.	MILITARY STANDARD MS27745
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 3 OF 7

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:50Z
Check the source to verify that this is the current version before use.

NOTES:
1. Terminal numbers do not appear on the relay header, there shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
2. Dimensions are in inches.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm) for three place decimals.
4. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
5. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this standard to the extent specified herein.
6. Relay is magnetically latched in both positions.
7. Caution note to observe polarity must appear on relays with dc coils.
8. Suppression level - the maximum induced transient voltage (back EMF) shall be 5 volts.

FIGURE 1. Outline drawing (for details see tables I and II) - Continued.

TABLE I. Relay characteristics.

MS part number MS27745-	Type	Coil type	Terminal type	Mounting or mating socket	Max weight in pounds
1	I	dc	solder	N/A	0.17
2	I	dc	plug-in	M12883/40-06,-12,-18,-24	0.17
3	I	ac	solder	N/A	0.17
4	I	ac	plug-in	M12883/40-06,-12,-18,-24	0.17
5	I	dc	plug-in	M12883/40-06,-12,-18,-24	0.17
6	I	dc	solder	N/A	0.17
7	I	ac	solder	N/A	0.17
8	I	dc	solder	N/A	0.17

APPROVED 3 JUNE 71 REVISED (G) FOR CHANGES SEE PAGES 1, 6, AND 7

User activities: Army - Navy - Air Force -
Review activities: Army - EC Navy - Air Force - 99
This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:50Z
Check the source to verify that this is the current version before use.

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

Review activities: Army -
Navy - EC
Air Force - 99

User activities: Army -
Navy -
Air Force -

TABLE II. Operating characteristics.

MS part no. MS27745	Coil data										Time-milliseconds-max				
	Coil	Nominal			Max		Max pick-up voltage 1/			Drop out voltage 1/	Operate	Release	Bounce		
		Volts	Freq Hz	Res Ω	Volts	Amp	Normal 1/	High temp test	Cont current test				Main		Aux
													NO	NC	NO NC
-1,-2,-6	X1, X2	28	dc	-	29	.075	18	19.8	22.5	-	15		1.0	1.0	- -
	Y1, Y2	28	dc	-	29	.075	18	19.8	22.5	-	15		1.0	1.0	- -
-3,-4,-7	X1, X2	115	400	-	122	.04	90	95.4	103.5	-	20		1.0	1.0	- -
	Y1, Y2	115	400	-	122	.04	90	95.4	103.5	-	20		1.0	1.0	- -
-5,-8	X1, X2	28	dc	-	29	.075	18	19.8	22.5	-	15		1.0	1.0	- -
	Y1, Y2	28	dc	-	29	.075	18	19.8	22.5	-	15		1.0	1.0	- -

1/ Over temperature range.

TABLE III. Rated contact load (amperes per pole) case grounded.

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 Phase				115/200 V ac, 3 Phase				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	100	10	10	-	-	10		-	-	10		-	-	
Inductive	100													
Inductive	20	8	8	-	-	8		-	-	8		-	-	
Motor	100	4	4	-	-	4		-	-	4		-	-	
Lamp	100	2	2	-	-	2		-	-	2		-	-	
Transfer, load														
Mechanical life reduced current	400	2.5	2.5	-	-	2.5		-	-	2.5		-	-	1/
Min current														

1/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:50Z
Check the source to verify that this is the current version before use.

ENVIRONMENTAL CHARACTERISTICS	-1, -2, -3, -4, and -5	-6, -7, and -8
Temperature range	-70°C to +125°C	-70°C to +125°C
Max altitude rating	300,000 ft	300,000 ft
Shock G-level	200 g's	100 g's
Duration	6 ms	6 ms
Max duration contact opening	10 µs	
Vibration-sinusoidal		
G-level	30 g's	20 g's
Frequency range	10 through 3000 Hz	10 through 3000 Hz
Vibration-random		
Applicable spec	MIL-STD-810	MIL-STD-810
Power spectral density	0.4 g^2/Hz	0.2 g^2/Hz
Acceleration	15 g's	15 g's

ELECTRICAL CHARACTERISTICS

Insulation resistance, initial . . . . . 100 megohms
After life or environmental tests . . . . . 50 megohms

Dielectric strength (sea level)	Initial	After life tests
Coil to coil and coil to case	1000 V rms	1000 V rms
Aux contacts		
All other points	1250 V rms	1000 V rms

Dielectric strength (altitude) 1/	80,000 ft	300,000 ft
Coil to coil and coil to case	350 V rms	500 V rms
Aux contacts		
All other points	350 V rms	500 V rms

Max contact drop initial		0.150 volt
After life test		0.175 volt
Overload current	40 amperes dc	60 amperes ac
Rupture current	50 amperes dc	80 amperes ac
Duty rating		Continuous

RFI specification
(Applicable to coil circuits of ac operated relays)

(G) Quality conformance inspection:
Performance of groups B and C inspection are not applicable.

(G) Supersession data: See table IV.

1/ Dielectric rating may be improved by suitable insulation of terminals and wiring after installation.

User activities: Army - Navy - Air Force -
Review activities: Army - Navy - EC Air Force - 99
This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

APPROVED 3 JUNE 71
REVISED (G) FOR CHANGES SEE PAGES 1, 6, AND 7

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:50Z
Check the source to verify that this is the current version before use.

Ⓖ Table IV. Supersession data. 1/

Superseded part number MS27745-	Superseding part number M6106/51-
1	001
2	002
3	007
4	008
5	005
6	003
7	009
8	006

1/ This table will also be used for logistic support.

User activities: Army - Navy - Air Force -
Review activities: Army - EC Navy - Air Force - 99
This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

APPROVED 3 JUNE 71 REVISED Ⓖ FOR CHANGES SEE SHEETS 1,6, AND 7

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:50Z
Check the source to verify that this is the current version before use.